Helping Others is Helping Yourself: How Chinese Foreign Aid Promotes China's Exports

by

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Abstract

We study how Chinese official foreign aid affects its exports to recipient countries. We first develop a trade model with foreign aid incorporating financial frictions and contractual incompleteness. Using data on Chinese aid and exports at multiple levels of granularity, we find that, consistent with the model's predictions, aid increases recipients' imports from both China and the rest of the world. The evidence also supports the mechanisms highlighted in the model: the export expansion is driven primarily by the alleviation of two-sided financial frictions on Chinese exporters and importers in recipient countries. Our results do not support alternative explanations.

Keywords: Chinese official foreign aid, exports, financial constraints, contract enforcement

JEL Codes: H87, F35, F14, P33

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1. Introduction

Over the past two decades, China has witnessed a dramatic increase in its official foreign aid. As illustrated in Figure 1, China's official foreign assistance increased from $5.45 billion in 2000 to $40.99 billion in 2021, reflecting an average annual growth rate of 22.1%.[1] By 2021, China had extended aid to a total of 146 countries. In recent years, its overseas assistance has even surpassed that of other major donors and lenders (Horn et al., 2021). This rapid growth has sparked considerable interest among researchers and policymakers regarding the economic consequences of China's foreign aid. Moreover, as a developing country, China offers a unique setting to examine how providing foreign assistance policy may affect the donor's own development.

Existing studies have focused primarily on the effects of Chinese official foreign aid on recipient countries, while paying much less attention to its implications for China itself or to the economic incentives underlying China's aid provision. This paper seeks to fill these gaps by investigating the domestic economic effects of China's foreign aid. Specifically, we examine how Chinese foreign aid shapes China's exports to recipient countries.[2] Over the past two decades, Chinese exports to aid-recipient countries also surged from $59.25 billion in 2000 to $971.21 billion in 2021, representing a remarkable 16-fold increase.[3] Moreover, as documented by Ding et al. (2018), despite China's relatively underdeveloped formal financial and legal institutions, Chinese exports have thrived in industries that rely heavily on external finance or contract enforcement.

This raises an important question: Is there a link between Chinese official foreign aid and China's exports to recipient countries, including their industry composition? To motivate our analysis, we first develop a three-country trade framework comprising China, a recipient country, and the rest of the world (ROW). This framework features heterogeneous sectors that differ in their dependence on external finance and their reliance on relationship-specific inputs. Chinese official aid influences trade patterns through two financial channels. First, aid acts as a form of sovereign endorsement and enhances the creditworthiness of Chinese exporters through government-backed financing arrangements. Second, aid generates positive spillovers to the recipient country's domestic financial infrastructure, raising the aggregate enforceability of contracts and relaxing financing constraints for all importers in the recipient country,

[1] Both numbers are in 2021 constant US dollars. They are obtained from the AidData 3.0 database (Custer et al. 2023).

[2] We use recipient countries' imports from China and China's exports to recipient countries interchangeably.

[3] By contrast, Chinese exports to non-recipient countries grew more modestly, from $353.22 billion to $1447.59 billion over the same period, roughly a fourfold increase. All numbers are in 2021 constant US dollars.

regardless of their products of origin. In the presence of financial frictions and incomplete contracts, common features in recipient countries, these aid-induced improvements alleviate both the exporter-side and importer-side wedges in trade transactions. The model yields four predictions: (i) Chinese aid increases exports from China to the recipient country; (ii) this effect is more pronounced in sectors with higher external finance dependence or higher contract intensity; (iii) aid also stimulates the recipient country's imports from the ROW; and (iv) the spillover to ROW exports is stronger in finance-dependent sectors but invariant to contract intensity.

To test these predictions, we combine detailed project-level data on Chinese official foreign aid with Chinese export data at various levels of granularity. We employ two complementary empirical strategies to carefully address the potential endogeneity concerns. First, we employ an instrumental variable (IV) regression approach that exploits time variation in aggregate Chinese aid supply shocks and cross-country differences in the likelihood of receiving Chinese funding (Nunn and Qian 2014; Dreher et al. 2021). We provide evidence supporting the relevance condition of the IV and conduct a series of exercises to bolster the plausibility of the exclusion restriction. Second, we follow the approach of Rajan and Zingales (1998) and provide direct evidence on the mechanisms emphasized in the model using more disaggregated data.

We begin with country-level analysis. The IV results indicate that, consistent with the model's predictions, Chinese official foreign aid significantly increases China's exports to recipient countries. This finding remains robust across alternative constructions of the IVs, different measures of Chinese official foreign aid, additional controls of international donors' aid, various sample selections, additional checks for the exclusion restriction of the IV, and a placebo test. Furthermore, the evidence also suggests that Chinese official aid not only increases recipient countries' imports from China but also boosts their imports from the ROW.

We next turn to more disaggregated data to provide additional evidence and shed light on the underlying mechanisms. The results confirm that Chinese official aid increases recipient imports from both China and the ROW at finer industry and product levels. Consistent with the theoretical mechanisms, the positive effect on Chinese exports is stronger in industries that are more dependent on external finance and more reliant on effective contract enforcement. In addition, aid stimulates exports from the ROW to recipient countries, particularly in finance-dependent industries.

Finally, we also conduct a series of additional tests to rule out competing explanations. The evidence suggests that other channels, including exports of infrastructure-related products

only, reductions in transportation costs or information frictions, or increased purchases by governments of recipient countries, are unlikely to account for our findings.

We contribute to the relevant literature in the following ways. First, it adds to the growing body of work on Chinese official foreign aid. Existing contributions in this field have primarily focused on the impacts on recipient countries, including economic growth (Dreher et al. 2021; Bluhm et al. 2025), finance (Bandiera and Tsiropoulos 2020), resource allocation (Dreher et al. 2019), environmental issues (Baehr et al., 2023), corruption (Isaksson and Kotsadam 2018), and stability (Gehring et al., 2022). Other research has explored the determinants of Chinese aid allocation, such as political alignment, natural resource endowments, and governance quality (Dreher and Fuchs 2015; Guillon and Mathonnat 2020). In contrast, we examine how Chinese foreign aid affects China's own economy, providing new evidence that aid can promote the donor country's exports by easing financial constraints and improving contract enforcement in recipient markets.

Second, our study relates to the broad literature on foreign aid. Existing research on aid and economic growth has produced mixed conclusions (e.g., Boone 1996; Burnside and Dollar 2000; Hansen and Tarp 2001; Easterly 2003). For instance, Burnside and Dollar (2000) argue that the effectiveness of aid depends on the recipient countries' institutions and policies, while Easterly (2003) shows that the idea that aid promotes growth in a favorable policy environment is not a robust empirical finding. Beyond growth, scholars have examined the effects of aid on conflict (Crost et al., 2014; Nunn and Qian 2014), corruption (Andersen et al., 2022), democratization (Birchler et al., 2016), and firm growth (Chauvet and Ehrhart 2018). Most of this literature focuses on aid from multilateral institutions such as the World Bank and IMF or from advanced economies such as the United States. By studying aid provided by a major developing country, our paper offers a complementary perspective and contributes to a more comprehensive understanding of the economic consequences of foreign assistance.

Finally, our study contributes to the literature on how government policies shape Chinese exports. Previous research has examined the effects of trade liberalization and related domestic reforms on firm productivity and export performance (e.g., Khandelwal et al., 2013; Brandt et al., 2017). Bai et al. (2014) further argue that a defining feature of China's informal institutional framework is the capacity of governments at different levels to leverage political and economic authority to influence firm behavior, particularly in export markets. Consistent with this view—and with broader evidence on the importance of domestic institutions in shaping export patterns (Levchenko 2007; Manova 2013; Nunn 2007; Ju and Wei 2010, 2011)—Ding et al. (2018) show that, despite relatively underdeveloped formal financial and legal institutions,

government policies in China play a pivotal role in shaping exporters' financial constraints and contract enforceability. Our paper provides new and complementary evidence that China's official foreign aid policy can promote exports, especially in industries that are more financially dependent and contract-intensive.

The remainder of this paper is organized as follows. Section 2 presents our conceptual framework. Section 3 describes our data and empirical strategy. Section 4 reports the country-level results. Section 5 provides evidence from disaggregated analyses and explores the mechanisms. Section 6 concludes.

2. A Simple Theoretical Model

In this section, we present a simple partial equilibrium model to guide our empirical analysis. We integrate foreign aid along with dependence on external finance and contract enforcement into a monopolistic competition model of trade.

2.1 Setup

The model comprises three countries: China (denoted by C), a recipient country (denoted by R) that receives Chinese aid, and the rest of the world (denoted by W). Each country consists of K sectors indexed by $k \in \{1, \dots, K\}$. In each sector, the final good aggregates heterogeneous varieties through a CES aggregator:

$$Q_k = \left(\int_{\omega \in \Omega_k} q(\omega)^{\frac{\sigma-1}{\sigma}} d\omega \right)^{\frac{\sigma}{\sigma-1}}, \quad (1)$$

where $\sigma > 1$ denotes the elasticity of substitution across varieties. Country R can source varieties $q(\omega)$ either domestically or through imports from China or the rest of the world.

The market operates under monopolistic competition. Each sector is characterized by two key parameters:

(1) *External finance dependence* ($d_k \in (0,1)$): the fraction of production (or trade-financing) costs that must be covered by external finance rather than internal funds.

(2) *Relationship specificity* ($\rho_k \in (0,1)$): the share of transaction value that is relationship-specific and thus subject to ex-post hold-up under incomplete contracts.

The supply-side structure follows the framework of Krugman (1980). In each country $i \in \{C, R, W\}$, there are N_i firms with homogenous productivity that sell to global markets. Free entry determines N_i endogenously through firms' profits across all markets. For the subsequent analysis, we focus on a partial equilibrium within a single sector k in country R,

suppressing the sector index for notational simplicity. General equilibrium variables—including wages w_i, the mass of active firms N_i, and real aggregate expenditure $E_R = Y_R P_R^{\sigma-1}$—are treated as exogenous.

2.2 Firm Problem

A producer in country $i \in \{C, R, W\}$ manufactures a single differentiated variety (suppressing the ω index) with quantity q_{iR} destined for the market in country R. Production follows a Leontief technology: $q_{iR} = \min\{m_{iR}^{RS}/\rho, m_{iR}^{NS}/(1-\rho)\}$, where m_{iR}^{RS} denotes relationship-specific inputs and m_{iR}^{NS} denotes non-relationship-specific (generic) inputs. The parameter $\rho \in (0,1)$ represents the input share for relationship-specific components and serves as a standard proxy for contract intensity.

Each producer is randomly paired with one domestic input supplier in country i and one foreign retailer in country R. Both suppliers and retailers operate in perfectly competitive sectors. The three parties engage in a sequential contracting process, structured as follows.

2.2.1 Timeline

The producer's optimization problem incorporates both financial frictions and contractual incompleteness. The timeline unfolds as follows:

(1) The producer makes a take-it-or-leave-it contract offer to the supplier, specifying an upfront transfer t_{iR} (which may be positive or negative). Due to contractual incompleteness, input quantities cannot be verified or contracted upon ex ante.

(2) The supplier accepts the offer (along the equilibrium path) and subsequently chooses input quantities. Given the Leontief technology, production requires inputs in fixed proportions, so the supplier optimally sets $m_{iR}^{RS}/m_{iR}^{NS} = \rho/(1-\rho)$. The unit cost for both input types is c_i.

(3) The producer and supplier bargain over the expected export revenue x_{iR}. Because relationship-specific inputs have zero salvage value outside the partnership, only the non-relationship-specific inputs can be redeployed. We assign property rights (the right of disposal) over these generic inputs to the producer. Consequently, the producer's outside option is $(1-\rho)c_i q_{iR}$, while the supplier's outside option is zero. The surplus (quasi-rent) available for division is therefore $s_{iR} = x_{iR} - (1-\rho)c_i q_{iR}$.

(4) The retailer makes a payment of x_{iR} to the producer, and the producer and supplier split it according to the bargaining outcome.

(5) The producer ships the goods to the retailer who sells them to the consumers. Due to

iceberg trade costs $\tau_{iR} > 1$, only q_{iR}/τ_{iR} units arrive in country R. The demand function is given by $q_{iR}/\tau_{iR} = E_R p_{iR}^{-\sigma}$ where p_{iR} denotes the consumer price. The retailer's total sales revenue is then $r_{iR} = p_{iR} q_{iR}/\tau_{iR} = E_R^{\frac{1}{\sigma}} \tau_{iR}^{\frac{1-\sigma}{\sigma}} q_{iR}^{\frac{\sigma-1}{\sigma}}$.

2.2.2 Financial frictions

The model incorporates two-sided financial frictions existing in both the exporting country i and the importing country R. We adopt the standard imperfect-enforcement framework to model the financial friction (Antràs and Foley 2015; Schmidt-Eisenlohr 2013; Manova 2013).

(1) *Importer-side friction*. A fraction $(1-d) \in (0,1)$ of the retailer payment x_{iR} is financed using its internal funds, while the remaining fraction d must be raised externally via trade-finance instruments (e.g., letters of credit, banker's acceptances, or supplier credit). Lenders in country R can enforce only a fraction $\kappa_R \in (0,1)$ of the promised repayment, reflecting the institutional quality and financial development of the destination market (e.g., creditor protection, collateral recovery efficiency, monitoring capacity, and payment-system reliability). Normalizing the risk-free interest rate to zero, the retailer must pledge a repayment amount l_{iR} such that the lender's expected recovery $\kappa_R l_{iR}$ covers the external financing need dx_{iR}. Thus, the participation constraint requires $\kappa_R l_{iR} \geq dx_{iR}$, yielding an equilibrium pledge of $l_{iR} = dx_{iR}/\kappa_R$. The retailer's total outlay for the export shipment therefore becomes $(1-d)x_{iR} + dx_{iR}/\kappa_R = (1-d+d/\kappa_R)x_{iR}$. Given perfect competition in the retail sector, the zero-profit condition implies:

$$x_{iR} = \frac{1}{1-d+\frac{d}{\kappa_R}} r_{iR} = \Lambda_R^I r_{iR}, \tag{2}$$

where $\Lambda_R^I \in (0,1)$ represents the importer-side financial wedge. The associated deadweight loss, $(1-\Lambda_R^I)r_{iR}$, arises because the gap between the contractual repayment and the lender's enforceable claim is dissipated as a real resource cost rather than redistributed. Two equivalent micro-foundations justify this dissipation: (i) *Costly state verification.* Following Townsend (1979), imperfect enforcement is modeled as a verification and recovery cost. A fraction of the pledged value is consumed by legal proceedings, administrative overhead, collateral liquidation discounts, and monitoring expenditures. (ii) *Third-party intermediation fees.* Alternatively, the wedge accrues to specialized trade-finance providers—confirming banks, trade-credit insurers, escrow agents, or export-credit agencies—that bridge the enforcement gap. From the importer's balance-sheet perspective, these are intermediation fees that scale inversely with domestic financial development; from the model's perspective, they are

permanent leakages that exit the supply chain's revenue stream.

(2) *Exporter-side friction.* The production cost $c_i q_{iR}$ in the exporting country is similarly subject to external financing constraints. A fraction d of total input costs must be raised ex ante through external funding, while the remaining $(1-d)$ is covered by the supplier's internal fund. The effective pledgeability rate is denoted by $\theta_{iR} \in (0,1)$, which captures lenders' assessment of the creditworthiness and institutional quality of the export destination. Following a similar derivation, the total cost incurred in production then equals $(1 - d + d/\theta_{iR})c_i q_{iR} = c_i q_{iR}/\Lambda_{iR}^E$, where $\Lambda_{iR}^E \in (0,1)$ denotes the exporter-side financial wedge. Note that this wedge operates as a sunk cost by the time the producer and supplier engage in ex-post Nash bargaining. Because external financing and input procurement strictly precede the bargaining stage, the incurred financing cost $c_i q_{iR}(1/\Lambda_{iR}^E - 1)$ cannot be recovered or renegotiated.

2.3 Equilibrium

We solve the partial equilibrium via backward induction. The retailer collects gross sales revenue r_{iR} and pays the trade value $x_{iR} = \Lambda_R^I r_{iR}$ to the producer. We assume that the relative bargaining power of the producer is $\beta \in (0,1)$. So, the producer gets $\beta s_{iR} + (1 - \rho)c_i q_{iR}$, and the supplier gets $(1-\beta)s_{iR} = (1-\beta)x_{iR} - (1-\beta)(1-\rho)c_i q_{iR}$. At the production stage, the supplier solves

$$\max_{q_{iR}} \ \pi_{iR}^S = (1-\beta)x_{iR} - (1-\beta)(1-\rho)c_i q_{iR} - \frac{1}{\Lambda_{iR}^E} c_i q_{iR}. \tag{3}$$

The first-order condition yields

$$q_{iR} = \frac{1}{c_i}\frac{\sigma-1}{\sigma}\Lambda_R^I \frac{1-\beta}{\frac{1}{\Lambda_{iR}^E}+(1-\beta)(1-\rho)} r_{iR} = \frac{1}{c_i}\frac{\sigma-1}{\sigma}\Lambda_R^I \tilde{\Lambda}_{iR}^E r_{iR}, \tag{4}$$

where $\tilde{\Lambda}_{iR}^E < 1$ denotes the composite exporter-side wedge attributed to both financial frictions and contractual incompleteness. Substituting equation (4) into the demand function yields

$$r_{iR} = \left(\frac{\sigma-1}{\sigma}\right)^{\sigma-1} E_R \tau_{iR}^{1-\sigma} c_i^{1-\sigma} (\Lambda_R^I)^{\sigma-1} (\tilde{\Lambda}_{iR}^E)^{\sigma-1}. \tag{5}$$

At the initial stage, the producer lump-sum upfront transfer t_{iR} such that the supplier breaks even: $\pi_{iR}^S + t_{iR} = 0$. The total profits earned by the producer is $\pi_{iR}^P = x_{iR} - c_i q_{iR}/\Lambda_{iR}^E$.

The sectoral aggregate exports from country i to country R is

$$X_{iR} = N_i x_{iR} = N_i \Lambda_R^I r_{iR}. \tag{6}$$

2.4 The effects of Chinese aid on exports

Chinese aid to country R, denoted by A_{CR}, alleviates the financial frictions for exporters in China that sell to R by enhancing the creditworthiness through government endorsement. This mechanism operates through several channels: Chinese aid projects often involve state-owned banks (such as China Development Bank or Export-Import Bank of China) that provide concessional loans tied to the use of Chinese contractors and suppliers. When a recipient country accepts Chinese aid, it signals government commitment to the project, reducing sovereign risk and improving the perceived creditworthiness of Chinese firms involved. Lenders view aid-backed projects as lower risk due to implicit government guarantees and the political cost of default on Chinese sovereign lending. Furthermore, aid agreements often include escrow accounts or direct payment mechanisms that ensure Chinese exporters receive payment, effectively increasing the collateralizable portion of their receivables. Therefore, we model θ_{CR} as an increasing function of aid:

$$\theta_{CR} = \theta(A_{CR}), \quad \text{with } \theta'(A_{CR}) > 0. \tag{7}$$

whereas θ_{RR} and θ_{WR} remain constant for domestic producers in country R and rest-of-world exporters, respectively.

Moreover, beyond the exporter-side effects, Chinese aid generates positive spillovers that deepen country R's domestic financial infrastructure. Aid inflows are channeled through recipient countries' local banking systems, modernize payment networks, and establish counterpart funds, collectively enhancing the enforcement capacity and credit-intermediation efficiency of domestic financial institutions. This economy-wide improvement raises the aggregate pledgeability rate κ_R, which relaxes the external financing constraint for all importers in country R. Crucially, this channel operates symmetrically across all trading partners, as it relaxes all importers' financial constraints in the recipient country. Accordingly, we specify:

$$\kappa_R = \kappa(A_{CR}), \quad \text{with } \kappa'(A_{CR}) > 0. \tag{8}$$

Now we consider an increase in Chinese aid to country R, $\Delta A_{CR} > 0$. This leads to $\Delta\theta_{CR} = \theta'(A_{CR})\Delta A_{CR} > 0$ and $\Delta\kappa_R = \kappa'(A_{CR})\Delta A_{CR} > 0$. The resulting change in China-to-R exports is

$$\Delta \ln X_{CR} = (\sigma - 1)\Delta \ln \tilde{\Lambda}^E_{CR} + \sigma \Delta \ln \Lambda^I_R = (\sigma - 1)\frac{\partial \ln \tilde{\Lambda}^E_{CR}}{\partial \theta_{CR}}\Delta\theta_{CR} + \sigma \frac{\partial \ln \Lambda^I_R}{\partial \kappa_R}\Delta\kappa_R, \tag{9}$$

Where

$$\frac{\partial \ln \tilde{\Lambda}_{CR}^{E}}{\partial \theta_{CR}} = \frac{1}{\theta_{CR}^{2}} \frac{d}{1-d+\frac{d}{\theta_{CR}}+(1-\beta)(1-\rho)} > 0. \tag{10}$$

$$\frac{\partial \ln \Lambda_{R}^{I}}{\partial \kappa_{R}} = \frac{1}{\kappa_{R}^{2}} \frac{d}{1-d+\frac{d}{\kappa_{R}}} > 0. \tag{11}$$

Thus, we have $\Delta \ln X_{CR} > 0$. This establishes the following proposition:

Proposition 1. An increase in Chinese aid to country R increases China-to-R exports.

Next, we examine how this response varies with external finance dependence d and contract intensity ρ. Differentiating equations (10) and (11) with respect to d and ρ gives:

$$\frac{\partial^{2} \ln \tilde{\Lambda}_{CR}^{E}}{\partial \theta_{CR} \partial d} = \frac{1}{\theta_{CR}^{2}} \frac{1+(1-\beta)(1-\rho)}{\left(1-d+\frac{d}{\theta_{CR}}+(1-\beta)(1-\rho)\right)^{2}} > 0, \tag{12}$$

$$\frac{\partial^{2} \ln \tilde{\Lambda}_{CR}^{E}}{\partial \theta_{CR} \partial \rho} = \frac{1}{\theta_{CR}^{2}} \frac{d(1-\beta)}{\left(1-d+\frac{d}{\theta_{CR}}+(1-\beta)(1-\rho)\right)^{2}} > 0, \tag{13}$$

$$\frac{\partial^{2} \ln \Lambda_{R}^{I}}{\partial \kappa_{R} \partial d} = \frac{1}{\kappa_{R}^{2}} \frac{1}{\left(1-d+\frac{d}{\kappa_{R}}\right)^{2}} > 0, \tag{14}$$

$$\frac{\partial^{2} \ln \Lambda_{R}^{I}}{\partial \kappa_{R} \partial \rho} = 0. \tag{15}$$

Therefore, cross-sector heterogeneity in the export change is:

$$\frac{\partial \Delta \ln X_{CR}}{\partial d} = (\sigma - 1) \frac{\partial^{2} \ln \tilde{\Lambda}_{CR}^{E}}{\partial \theta_{CR} \partial d} \Delta \theta_{CR} + \sigma \frac{\partial^{2} \ln \Lambda_{R}^{I}}{\partial \kappa_{R} \partial d} \Delta \kappa_{R} > 0 \tag{16}$$

$$\frac{\partial \Delta \ln X_{CR}}{\partial \rho} = (\sigma - 1) \frac{\partial^{2} \ln \tilde{\Lambda}_{CR}^{E}}{\partial \theta_{CR} \partial \rho} \Delta \theta_{CR} + \sigma \frac{\partial^{2} \ln \Lambda_{R}^{I}}{\partial \kappa_{R} \partial \rho} \Delta \kappa_{R} > 0 \tag{17}$$

This establishes the following proposition:

Proposition 2. The positive effect of Chinese aid on China-to-Rexports is stronger for sectors with higher external financial dependence and contract intensity.

We now examine the implications for rest-of-world exports to country R. As $\Delta \theta_{WR} = 0$, the export change is given by

$$\Delta \ln X_{WR} = \sigma \Delta \ln \Lambda_{R}^{I} = \sigma \frac{\partial \ln \Lambda_{R}^{I}}{\partial \kappa_{R}} \Delta \kappa_{R} \tag{18}$$

This yields the following result:

Proposition 3. An increase in Chinese aid to country R raises rest-of-world exports to country R.

Sectoral heterogeneity in this response is characterized by:

$$\frac{\partial \Delta \ln X_{WR}}{\partial d} = \sigma \frac{\partial^2 \ln \Lambda_R^I}{\partial \kappa_R \partial d} \Delta \kappa_R > 0, \tag{19}$$

$$\frac{\partial \Delta \ln X_{WR}}{\partial \rho} = \sigma \frac{\partial^2 \ln \Lambda_R^I}{\partial \kappa_R \partial \rho} \Delta \kappa_R = 0. \tag{20}$$

This leads to the following proposition:

Proposition 4. The effect of Chinese aid on country R's imports from the ROW is larger in sectors with greater external finance dependence, but invariant to relationship specificity.

3. Data and Empirical Strategy

3.1 Chinese official foreign aid

We obtain data on Chinese foreign aid from AidData's Global Chinese Development Finance Dataset, Version 3.0. This dataset offers comprehensive coverage of 20,985 official development programs initiated by China, amounting to $1.34 trillion and spanning 154 countries or regions between 2000 and 2021. For each project, AidData 3.0 provides detailed information, including the recipient country or region, the year of the commitment, the financial value expressed (in constant 2021 US dollars), and the project's intent or purpose. Following Dreher et al. (2021), we refine the sample in several ways. We exclude projects identified only at the regional level (e.g., Africa or Asia) rather than for a specific country, as well as multi-country projects. We also remove projects that were double-counted or subsequently canceled. In addition, we exclude COVID-19–related projects, including those involving information dissemination, testing, prevention, immunization, or medical supplies.

After applying these restrictions, our final sample consists of 15,906 projects. We use these data to construct measures of Chinese official foreign aid at the recipient country–year level. Our main measure is the annual financial value of Chinese official aid to each recipient country in 2021 constant dollar. This serves as our primary independent variable. As an alternative measure for robustness checks, we also calculate the number of projects received by each country in a given year.[4] Both variables are expressed in natural logarithms. Figure 2 presents the global distribution of Chinese official aid across recipient countries. Panel A depicts the distribution of the total aid value received by these countries from 2000 to 2021, while Panel B shows the distribution of the number of Chinese official finance projects. In both panels, darker shading indicates higher levels of Chinese government financing. Notably, the

[4] Since the AidData only provides the commitment year of aid, the exact time of the aid disbursement is not available. We assume a one-year lag in the construction of these two measures of Chinese official foreign aid.

distribution patterns of these two variables closely align with a correlation coefficient of 0.86. As illustrated in the figure, African countries and Russia are among the largest recipients of Chinese government finance.

[Figure 2 is about here]

3.2 Chinese exports and control variables

Our main dependent variable is the value of Chinese exports to recipient countries, measured in constant 2021 US dollars and expressed in natural logarithm. We obtain data on recipient countries' imports from China at the country–product–year level from the UN COMTRADE database, with products defined at the Harmonized System (HS) 6-digit level.

We aggregate these data in two ways. First, we collapse them to the country–year level to examine how Chinese official finance affects a recipient's total imports from China. Second, to explore industry heterogeneity, we aggregate exports to the country–industry level using either ISIC 3-digit or SITC 4-digit classifications. This aggregation is implemented using concordance tables from the World Integrated Trade Solution (WITS). We exclude developed countries from the main sample, as none received Chinese aid during the sample period.[5] The final dataset therefore covers Chinese exports—at multiple levels of disaggregation—and foreign aid flows to 149 non-developed trading partners from 2000 to 2021.

In the empirical analysis, we control for country-level characteristics that may influence imports from China. Our primary controls include real GDP and population (both in natural logarithms), real GDP growth, and the bilateral exchange rate between the Chinese renminbi (RMB) and the recipient country's currency. These variables are obtained from the World Bank's World Development Indicators (WDI) and the United Nations Statistics Division. We also control for the logarithm of the average tariff rate imposed on Chinese exports by each recipient country, sourced from WITS. Table 1 reports summary statistics for the key country-level variables used in the analysis, and Online Appendix B provides detailed variable definitions and data sources.

[Table 1 is about here]

3.3 Empirical specifications

3.3.1 Country-level analysis

To estimate the effect of Chinese official finance on Chinese exports to recipient countries,

[5] Including developed countries in the sample does not affect our findings.

we begin with the following country-level specification:

$$\ln(Export_{ct}) = \beta_0 + \beta_1 OF_{ct} + \beta_x X_{ct-1} + \varphi_t + \varphi_c + \epsilon_{ct} \quad (21)$$

where c and t represent recipient country and year, respectively. The dependent variable, $\ln(Export_{ct})$, denotes the log value of China's export to the recipient country. The key explanatory variable, OF_{ct}, measures Chinese official finance and is defined as either the log value or the log of number of projects received by the country in a given year.[6] The vector X_{ct-1} includes a set of control variables at the recipient country level, including log real GDP, real GDP growth, log population size, the exchange rate, and the log of average tariff rate. All control variables are lagged by one period. We also include year fixed effect, φ_t, and country fixed effect, φ_c, to control for the influence of time-varying shocks that are common across countries and the impacts of time-invariant country-specific factors.

A central concern is the potential endogeneity of Chinese foreign aid. On the one hand, China may allocate more aid to countries where current exports are low in order to foster future trade relationships. On the other hand, China may direct more aid to its major export destinations. To address this issue, we adopt two complementary strategies.

First, we implement an IV approach following Dreher et al. (2021) and Nunn and Qian (2014). This IV strategy utilizes the interaction of two components, expressed as $IV_{ct} = P_c \times X_{t-1}$. The first term captures the probability of a country receiving Chinese official finance during the sample period. In our baseline case, we follow Dreher et al. (2021) and calculate this probability as the fraction of years between 2000 and 2021 in which the recipient country received at least one Chinese project commitment. For the sake of robustness, we also compute two alternative measures of the average probability of a country receiving Chinese official aid: one using data from the first decade of our sample period (2000-2010) and the other from the pre-sample period (1995-1999).

The second term proxies for supply-side shocks associated with Chinese official foreign aid. In line with the methodologies used by Dreher et al. (2021) and Bluhm et al. (2025), we use production levels of six strategic input materials—iron, steel, timber, aluminum, cement, and glass—as a proxy for this supply shock.[7] These materials are considered strategic products that are less likely to be influenced by demand shocks and are essential for infrastructure

[6] Since there are country-year observations with zero aid or exports, we add one to the aid and export variables before taking logarithm.

[7] Dreher et al. (2021) also use foreign reserve holdings as an alternative proxy of the supply shock of Chinese foreign aid. We do not use it in our main analysis as it is likely to be endogenous in our case. Nevertheless, our main results remain robust to the use foreign research holdings.

projects often linked to Chinese government aid. Higher production levels may increase the incentive to channel aid toward projects that absorb surplus capacity. Given that the production of these materials tends to trend over time, we first detrend the data and then employ principal component analysis to extract a single input factor index.

Figure 3 plots the relationship between the endogenous variable and the baseline IV to assess instrument relevance. Panels A and B assess Chinese foreign aid using the aid value and the number of projects, respectively. In both cases, we observe a clear positive correlation, supporting the relevance condition.

[Figure 3 is about here]

The exclusion restriction requires that production of the six inputs does not directly affect China's exports to recipient countries. We take several steps to support this assumption. First, in all regressions, we exclude exports of the six input materials from the construction of the dependent variable. Second, to address potential indirect effects, we exclude exports from downstream sectors that use these materials as primary inputs. Next, we further exclude sectors that rely on the outputs of those downstream industries as their main inputs, thereby accounting for possible second-order indirect effects. Finally, to address the concern that our IV may potentially capture a common business cycle factor that drives both China's production of the six materials and its overall supply capacity and exports, we control for China's GDP growth interacted with recipient-country fixed effects in the IV regression as a further check.

3.3.2 Country-industry (product) level analysis

Our second identification strategy follows the approach of Rajan and Zingales (1998), who emphasize that direct evidence on the underlying mechanisms can serve as a "smoking gun" for causality. Using more disaggregated country–industry data, we test whether Chinese foreign aid disproportionately promotes exports in sectors that rely more heavily on external finance or contract enforcement, as predicted by our model. We also examine whether aid disproportionately increases recipient imports from the ROW in finance-dependent sectors. In addition, we offer micro-level evidence to rule out alternative explanations for the observed export expansion. The direct evidence on the mechanisms can thus offer more confidence in causality.

Before examining the underlying mechanisms, we first verify that the baseline results hold at a more disaggregated level by estimating:

$$\ln(Export_{ict}) = \beta_0 + \beta_1 OF_{ct} + +\mu X_{ct-1} + \rho tariff_{cit-1} + \alpha_{ic} + \gamma_t + \varepsilon_{ict} \quad (22)$$

where *i, c* and *t* represent industry (or product), recipient country and year, respectively. We also include tariffs, $tariff_{cit-1}$, and country-industry fixed effects α_{ic}, as additional controls in this specification.

To examine the underlying mechanisms, we exploit cross-industry variation in reliance on external finance and contract enforcement and consider the following country-industry-year level regression:

$$\ln(Export_{ict}) = \beta_0 + \beta_1 OF_{ct} \times financedep_i + \beta_2 OF_{ct} \times contract_i$$

$$+\mu X_{ct-1} + \rho tariff_{cit-1} + \alpha_{ic} + \gamma_{ct} + \varepsilon_{ict} \quad (23)$$

where $financedep_i$ is a dummy variable equal to one if an industry's external finance dependence exceeds the sample mean. $contract_i$ is a dummy variable equal to one if an industry's contract intensity exceeds the sample mean. The coefficients of interest are β_1 and β_2. Positive and significant estimates would indicate that Chinese aid disproportionately increases exports in industries that rely more on external finance or contract enforcement, consistent with the theoretical framework. Compared to equation (22), this specification includes the more stringent country-year fixed effects, γ_{ct}, which absorb all time-varying country-level factors that may influence the outcome variable.[8] As a robustness check, we also consider including country-industry time trends in equations (22) and (23) as additional controls.

4. Evidence from Country-Level Data

This section presents our country-level evidence regarding the effects of Chinese official foreign aid on its exports to recipient countries. We begin with baseline IV regression results and then report a range of robustness checks and extensions.

4.1 Baseline IV regression results

Before presenting the formal regression results, we first conduct a visual inspection of the relationship between these two variables. Figure 4 presents binned scatter plots illustrating the association between China's exports to recipient countries and Chinese official aid, controlling for year fixed effects. Panel A uses the log value of aid, while Panel B uses the log number of projects. In both panels, we observe a clear positive correlation between Chinese exports and

[8] The inclusion of country-year fixed effects absorbs the impact of aid. Similarly, the inclusion of country-industry level fixed effects absorbs the effects of the industry-level measure of external finance dependence and contract enforcement.

official foreign aid.

[Figure 4 is about here.]

To better establish a causal relationship, we implement the IV strategy described in Section II.C and estimate the model using two-stage least squares (2SLS). Column (1) of Table 2 reports the baseline IV regression results. In columns (2) and (3), we consider two alternative instruments that employ different sample periods to compute the probability of receiving Chinese aid. Across all specifications, the coefficients on Chinese official aid are positive and statistically significant. The estimated effect is also economically meaningful. For example, the results in column (1) suggest that a 1% increase in Chinese official aid is associated with roughly a 0.1% increase in China's export value to the recipient country.

Table 2 also reports the first-stage results. The estimated coefficients on the IV are positive and statistically significant at the 1% level across all columns, confirming a strong relationship between the IV and the endogenous variable. Moreover, the Kleibergen-Paap rk Wald F statistics indicate that our IVs pass the weak identification test.

[Table 2 is about here]

4.2 Robustness checks

In this subsection, we conduct a series of sensitivity analyses to assess the robustness of our baseline findings. First, we examine whether the results depend on how Chinese official aid is measured. In Column (1) of Table 3, we re-estimate the baseline IV specification using the log number of aid projects instead of aid value. The estimated coefficient remains positive and statistically significant.

Second, AidData classifies Chinese official finance into nine categories: loans, technical assistance, debt forgiveness, debt rescheduling, grants, scholarships, export buyer's credit, supplier's credit, and a residual "vague" category. To ensure that our findings are not driven solely by buyer's or supplier's credits, we exclude these two categories when constructing the aid measure. Column (2) of Table 3 shows that the results are unchanged.

Third, we control for aid from other major international donors. Specifically, we include the log value of aid from OECD Development Assistance Committee (DAC) members, the World Bank (IBRD-IDA), and the United Nations. As shown in Column (3), the coefficient on Chinese official aid remains positive and significant, suggesting that our results are not confounded by other donors' activities.

Next, to address concerns about outliers, we trim observations below the 1st percentile and above the 99th percentile of the distributions of the dependent variable, the key

independent variable, and control variables. Column (4) shows that the main result is unaffected. In our main analysis, we exclude developed countries as they are unlikely to receive any official aid from China. While not reported, we also conduct a robustness check by adding developed countries in the sample. The estimated effect of aid remains robust in this extended sample.

We also further probe the exclusion restriction underlying our IV. In all regressions, we exclude exports of the six strategic input materials used to construct the instrument. However, supply shocks to these materials could indirectly affect exports in downstream industries. To address this concern, Column (5) excludes exports from sectors that use these six materials as primary inputs. Since the 2007 China Input-Output (IO) table does not provide separate sectors for these six inputs, we identify the relevant downstream sectors based on their sector descriptions. For example, sector 032 ("Wood Processing and Wood, Bamboo, Rattan, Palm, and Straw Products") is described as "processing raw wood into wood products," and is therefore excluded. Using this approach, we identify a total of six sectors (204 products), including sector 032, sectors 033 (Furniture Manufacturing), 050 (Cement, Lime, and Gypsum Manufacturing), 051 (Cement and Gypsum Products Manufacturing), 053 (Glass and Glass Products Manufacturing), and 063 (Metal Products Manufacturing). Excluding these sectors does not alter our findings.

As a further check, column (6) accounts for potential second-order indirect effects. For each of the six downstream sectors excluded in column (5), we identify the top five sectors that use their outputs as primary inputs based on the 2007 China IO table and exclude them as well (729 products in total). Even after excluding both direct and second-order downstream sectors, the estimated effect of Chinese aid on exports remains positive and statistically significant.

Another potential concern regarding the exclusion restriction is that our instrument may capture a common business cycle factor that drives both Chinese aid supply and its export performance. To mitigate this concern, we augment the IV regression by including the interaction of China's GDP growth rate with recipient-country fixed effects as additional controls, thereby absorbing cyclical covariation that is common across recipient countries. The results, reported in column (7) of Table 3, indicate that our core findings hold strongly.

[Table 3 is about here]

Finally, to address the potential influence of unobserved factors on our results, we conduct a placebo test in which we randomly reassign Chinese official aid across various countries and years. We then estimate our baseline IV regression using this randomized data. If unobserved factors were indeed driving our results, we would expect to see a significantly positive effect

of the randomly reassigned Chinese official finance on export values in this artificial dataset. Conversely, an insignificant coefficient in the placebo test would suggest that the estimation of China's export values related to Chinese official foreign aid is not biased by unobserved factors. We repeat the regression 1000 times and present the distribution of estimated coefficients in Figure 5. The median estimated coefficient on the aid variable in the placebo test is found to be -0.00003, with a standard deviation of 0.002. In contrast, our IV estimate reported in column (1) of Table 2 is 0.095. Our findings are unlikely driven by unobserved factors.

[Figure 5 is about here]

Overall, the extensive robustness checks consistently support the conclusion that Chinese official foreign aid significantly increases China's exports to recipient countries.

4.3 The effect on imports from the ROW

In this subsection, we examine further the effect of Chinese official aid on recipient countries' imports from the ROW. This analysis serves two main purposes. First, the findings carry important policy implications for recipient countries as well as for the ROW. Second, they allow us to test a key prediction of the model (Proposition 3), which posits that the positive spillovers from aid relax the financing constraints faced by all importers in the recipient country, thereby increasing imports not only from China but also from the ROW.

To pursue this, we gather data on each country's imports from all its trading partners using the UN COMTRADE database. This allows us to calculate the log value of a recipient country's total imports from the world, the log value of its imports from all other countries except China, and the share of imports from China relative to total imports. We then estimate equation (21) using these variables as our dependent variables.

The IV regression results are presented in columns (1) - (3) of Table 4. In column (1), we use a recipient country's total imports from the world as the dependent variable. The estimates show that Chinese official aid has a positive and statistically significant effect on overall import expansion. Column (2) focuses on the impact on the ROW only. Consistent with our model's prediction, aid is also found to have a significantly positive effect on imports from the ROW. Moreover, quantitatively, the estimated effect of aid on imports from ROW is much smaller than its effect on imports from China reported in column (1) of Table 2. This difference again aligns with our theoretical framework. In particular, it is consistent with the mechanism outlined in equations (9) and (18): aid stimulates imports from China by relaxing financial constraints on both the exporter and importer sides, whereas imports from the ROW are affected only through the relaxation of financial constraints on the importer side. The large

difference in magnitudes also suggest that exporter-side effect of the aid is quantitatively more important. Finally, column (3) uses the share of imports sourced from China as the dependent variable. The coefficient on the aid variable is positive and statistically significant, indicating a significant increase in China's share of total imports.

[Table 4 is about here]

Taken together, the results from country-level analysis suggest that Chinese official aid stimulates imports from both China and the ROW, resulting in an overall expansion of the recipient country's total imports. However, the effect is much stronger for imports from China than for those from the ROW, suggesting that China captures a disproportionate share of the trade growth associated with its aid.

5. Evidence from More Disaggregated Data and Mechanisms

5.1 The effect on country-industry level exports

Thus far, our analysis has shown, using country-level data, that Chinese aid increases recipients' aggregate imports from China. To provide additional supporting evidence and shed light on the underlying mechanisms, we now turn to more disaggregated data.

We begin by estimating equation (22) at the country–industry level. The results are reported in column (1) of Table 5. Panel A uses the log value of aid as the key explanatory variable, while Panel B uses the log number of aid projects. In column (2), we additionally control for country-industry time trends. Across both panels, the estimated coefficients on Chinese official aid are positive and statistically significant. These findings are in line with the country-level results and confirm that Chinese aid promotes exports at the recipient country–industry level.

5.2 Examining the underlying mechanisms

Next, we investigate the mechanisms proposed by our theoretical framework. Specifically, we test whether Chinese official aid promotes Chinese exports by relaxing financial constraints and improving contract enforcement in recipient countries. To do so, we estimate equation (23), which interacts aid with industry-level measures of external finance dependence and contract intensity. The results are presented in columns (3) and (4) of Table 5 with column (4) additionally controlling for country-industry time trends. In both columns, the coefficients on the interaction terms are positive and significantly, indicating that the export-promoting effect of aid is stronger in industries that rely more heavily on external finance or contract

enforcement.

Columns (5) - (8) further explore whether Chinese aid affects recipient countries' imports from the ROW, and whether this effect varies across industries. Columns (5) re-estimates equation (22) using the log of recipient imports from the ROW as the dependent variable. In column (6), we include country-industry time trends as additional controls. The results show that Chinese aid increases imports from other trading partners as well. Also, consistent with the country-level evidence, the estimated effect on imports from the ROW is quantitatively much small than that on imports from China.

Columns (7) and (8) introduce the interactions between aid and external finance dependence and contract intensity. The results suggest that the positive effect of aid on imports from the ROW is significantly stronger in industries relying more on external finance. This finding is consistent with the theoretical prediction that relaxed financial constraints on the importer side facilitate import activities from other trading partners. In contrast, we find no evidence that Chinese aid increases imports from the ROW through improved contract enforceability.

[Table 5 is about here]

The results in Table 5 provide strong support to our theoretical predictions. One remaining concern, however, is that sectors with high external finance dependence or contract intensity might happen to be those where China has a comparative advantage vis-à-vis recipient countries. This is unlikely to be the case as China's financial and legal institutions are also under-developed (e.g., Ding et al., 2018). To formally address this concern, we further control for normalized revealed comparative advantage (RCA, described in section IV.C.3 for details) at the country-industry-year level. Appendix Table A1 re-estimates equations (23) with an RCA dummy and its interaction with aid. In columns (2) and (4) of Table A1, we also additionally control for country-industry time trends. The inclusion of RCA or its interaction term with aid does not alter our main conclusions: the interaction terms between aid and external finance dependence, and between aid and contract intensity, remain positive and statistically significant. While RCA itself has a positive effect on Chinese exports, its interaction with aid is significantly negative, suggesting that the aid effect is not merely driven by comparative advantage.

5.3 Competing hypotheses

The results in Table 5 and Appendix Table A1 provide robust evidence that Chinese official aid promotes China's exports through the relaxation of financial constraints and the

improvement of contract enforceability in recipient countries. In this subsection, we present additional evidence to rule out several alternative explanations.

5.3.1 Infrastructure-related products

One alternative explanation is that the observed increase in Chinese exports is driven solely by infrastructure-related goods. Since Chinese aid projects are often linked to infrastructure and construction and these projects may favor procurement from Chinese firms, it is possible that aid boosts exports solely through the sourcing of infrastructure materials and construction inputs.

We evaluate this hypothesis in three ways. First, this explanation is not consistent with the finding that Chinese aid also increases recipient countries' imports from the ROW. Second, if imports of infrastructure goods are driving the results, the estimated aid effect should disappear once such products are excluded. In our baseline regressions, we already omit obvious infrastructure inputs (e.g., steel, iron, cement, timber, glass, and aluminum). To be more conservative, we further exclude products that serve as major inputs into construction. Using China's 2007 Input–Output table, we identify the five sectors that supply the largest share of inputs to the construction sector and drop all products mapped to these sectors.[9] Columns (1) – (3) of Table 6 report the results using country-level, country–industry-level, and country–product-level data, respectively. In all specifications, the coefficient on Chinese official aid remains positive and statistically significant, indicating that our findings are not driven solely by infrastructure-related products.

Third, we exploit product upstreamness to test whether the aid effect is concentrated in upstream (input-intensive) industries. Infrastructure-related goods tend to be upstream in the production chain. If procurement for aid projects explains our results, the aid effect should be stronger for upstream products. Using the upstreamness index from Chor et al. (2021), we classify products as upstream if their index exceeds the sample mean and downstream otherwise. Contrary to the procurement hypothesis, columns (4) and (5) of Table 6 show that the effect of aid is larger for downstream products. Taken together, the evidence in Table 6 suggests that the positive effect of Chinese official aid on imports from China cannot be attributed solely to infrastructure procurement.

[Table 6 is about here]

9 These five sectors include 059 (Steel Rolling and Processing Industry), 052 (Brick, Tile, stone, and Other Building Materials Manufacturing), 050 (Cement, lime, and Gypsum Manufacturing), 051 (Cement and Gypsum Products Manufacturing), and 063 (Metal Products Manufacturing).

5.3.2 Reduction of transportation costs

Given that Chinese official aid is often linked to infrastructure projects, another potential explanation for the rise in Chinese exports could be the reduction in transportation costs. To test this mechanism, we construct two product-level proxies for transportation costs. The first is the value-to-weight ratio, where higher values imply lower sensitivity to transport costs. The second is the cif-to-fob ratio, following Anderson and van Wincoop (2004), where higher values indicate greater transport cost sensitivity. We interact each measure with Chinese official aid and include the interaction term in equation (23). If reduced transportation costs were the primary channel, aid should have a weaker (stronger) effect on products with a higher value-to-weight (cif-to-fob) ratio.

Table 7 reports the IV results. Columns (1) and (2) include the interaction with the value-to-weight ratio, while columns (3) and (4) include the interaction with the cif-to-fob ratio. Columns (1) and (3) use our baseline specification with country–product and year fixed effects; Columns (2) and (4) add country–year fixed effects for a more stringent specification.[10] The results do not support the transportation cost channel. Although the coefficient on Chinese official aid remains positive and significant in Columns (1) and (3), the interaction terms are statistically insignificant across all specifications.

[Table 7 is about here]

5.3.3 Reduction of information frictions

The third possible explanation for the positive export effect is that Chinese official aid reduces information frictions, aiding potential Chinese exporters in gaining knowledge about the recipient countries' markets. To test this hypothesis, we conduct two sets of analyses in Table 8. First, using country-level data, we interact Chinese official aid with proxies for information barriers—namely geographic distance and linguistic proximity between China and the recipient country—and include the interaction terms in equation (21). If aid mitigates information frictions, its effect should be stronger where information barriers are higher. However, as shown in columns (1) and (2), the interaction terms are statistically insignificant, providing no support for this mechanism.

Second, we investigate whether aid disproportionately promotes exports in products where China has a comparative advantage. If aid alleviates information frictions, it should particularly boost exports in sectors where China is already relatively competitive. To assess

[10] The Chinese official aid variable submerges with the inclusion of country-year fixed effects.

this, we construct a normalized revealed comparative advantage (NRCA) index following Yu et al. (2009):

$$NRCA_{cht} = \frac{x_{cht}}{x_t} - \frac{x_{ht}x_{ct}}{(x_t)^2}$$

where x_{cht} represents country *c*'s exports value of product *h*. x_{ct} is country *c*'s total export value of all products in a given year, x_{ht} is the world's total export value of product *h* in a given year, and x_t denotes the world's total exports value in year *t*. The NRCA index ranges from -0.25 to 0.25, with 0 representing a comparative-neutral position. A higher positive NRCA value indicates that a country is relatively more competitive in exporting that particular product.

We define a relative comparative advantage (RCA) dummy equal to one if China's NRCA exceeds that of the recipient country, and zero otherwise. We then interact this dummy with Chinese official aid and include both terms in equation (22). The IV results, reported in columns (3) and (4) of Table 8, do not support the information-friction channel. The interaction coefficients are significantly negative, suggesting that aid has a stronger effect in sectors where China has lower relative comparative advantage in the recipient country.

[Table 8 is about here]

5.3.4 Government purchase

The last alternative mechanism we explore is government purchase. It is possible that aid may enhance China's influence over the government procurement decisions of recipient countries. This argument obviously is inconsistent with our finding that aid also stimulates the ROW's exports to recipient countries. To formally test this hypothesis, we adopt the approach of Berger et al. (2013) and investigate whether the impact of Chinese official finance on imports is more pronounced in countries with higher government expenditure shares or in industries with greater levels of government procurement and import activity. To do this, we include the interaction term between Chinese official aid and the recipient country's government expenditure-to-GDP ratio in equation (21) as an additional regressor. The IV regression results are reported in column (1) of Table 9. The results do not support the government purchase explanation. While the estimated coefficient on Chinese official aid remains significantly positive, the coefficient on the interaction term is negative and statistically insignificant.

We also utilize SITC 2-digit industry-level measures of government purchases and imports, as constructed by Berger et al. (2013). Industries are categorized as having high or low government purchase (import) shares based on whether their respective shares exceed the mean value across all industries. We then estimate equation (22) by incorporating the

interaction term between Chinese official aid and the industry-level measures of government purchases. Since the industry-level measures are available at the SITC 2-digit level, we aggregate our data accordingly to the country-SITC-2-digit level.

Columns (2) – (5) of Table 9 report the results. Columns (2) and (3) use the high government purchase share dummy, while columns (4) and (5) use the high government import share dummy. Columns (2) and (4) include country–industry and year fixed effects; columns (3) and (5) add country–year fixed effects. Across all specifications, the interaction terms are statistically insignificant.

[Table 9 is about here]

Overall, the evidence in Tables 6 – 9 indicates that our main findings cannot be explained by infrastructure procurement, transportation cost reductions, information frictions, or government procurement channels. Instead, the results are most consistent with the financial constraint and contract enforcement mechanisms emphasized in our theoretical framework.

6. Conclusions

China is playing an increasingly significant role in international official finance, yet there is still no clear consensus on the nature and consequences of its overseas assistance. In particular, relatively little is known about how Chinese official foreign aid affects China's own economic development. Our paper aims to fill this research gap by examining the effects of official foreign aid on China's trade patterns, thereby contributing to a better understanding of the economic implications of China's foreign assistance policy.

We first develop a simple theoretical framework illustrating how Chinese official aid can stimulate exports not only from China but also from the ROW to recipient countries. We then test the model's predictions using detailed project-level data from AidData combined with export data at multiple levels of aggregation. Employing an IV strategy, we find that Chinese official foreign aid significantly promotes China's exports to recipient countries. Moreover, the results indicate that aid also increases the ROW's exports to the recipient countries.

To uncover the underlying mechanisms, we provide evidence that Chinese official foreign aid alleviates financial constraints and improves contract enforceability in recipient countries, leading to a more pronounced effect on exports in industries that rely heavily on external finance and contract enforcement. We also provide additional evidence to rule out other competing explanations.

Overall, our findings highlight that China's official foreign aid is an effective instrument for expanding its own exports. By relaxing financial constraints and strengthening contract

enforceability in recipient countries, Chinese aid generates mutual economic benefits. These results contribute to a more nuanced understanding of economic implications of China's growing role in international official finance.

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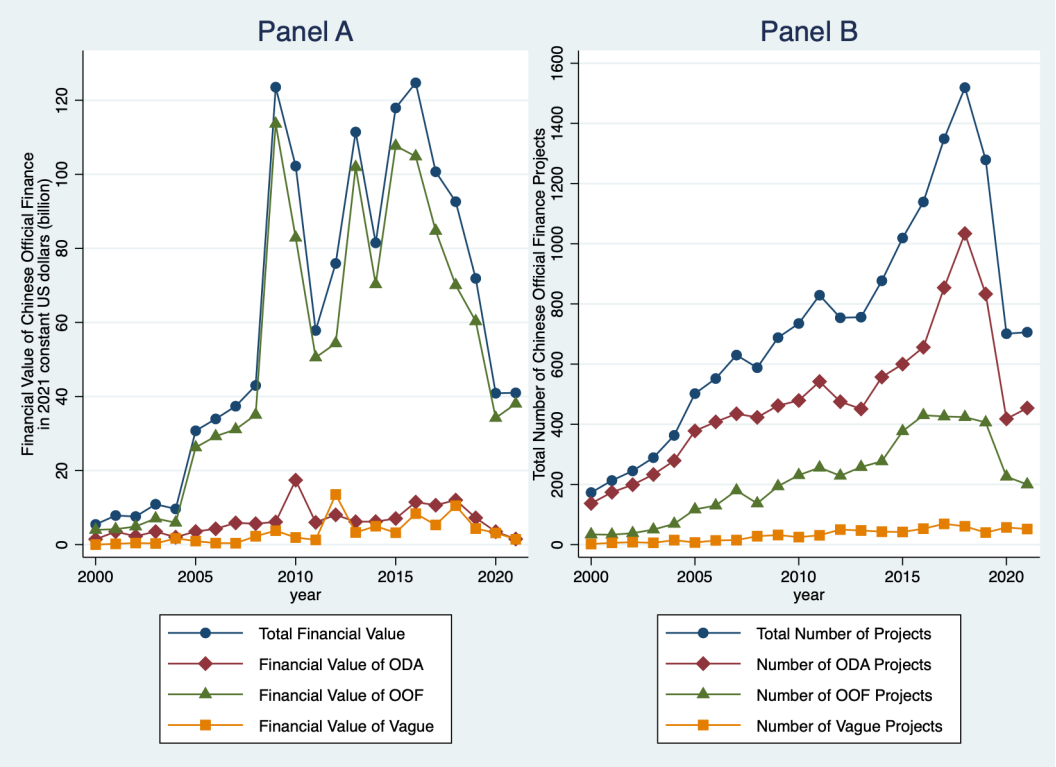
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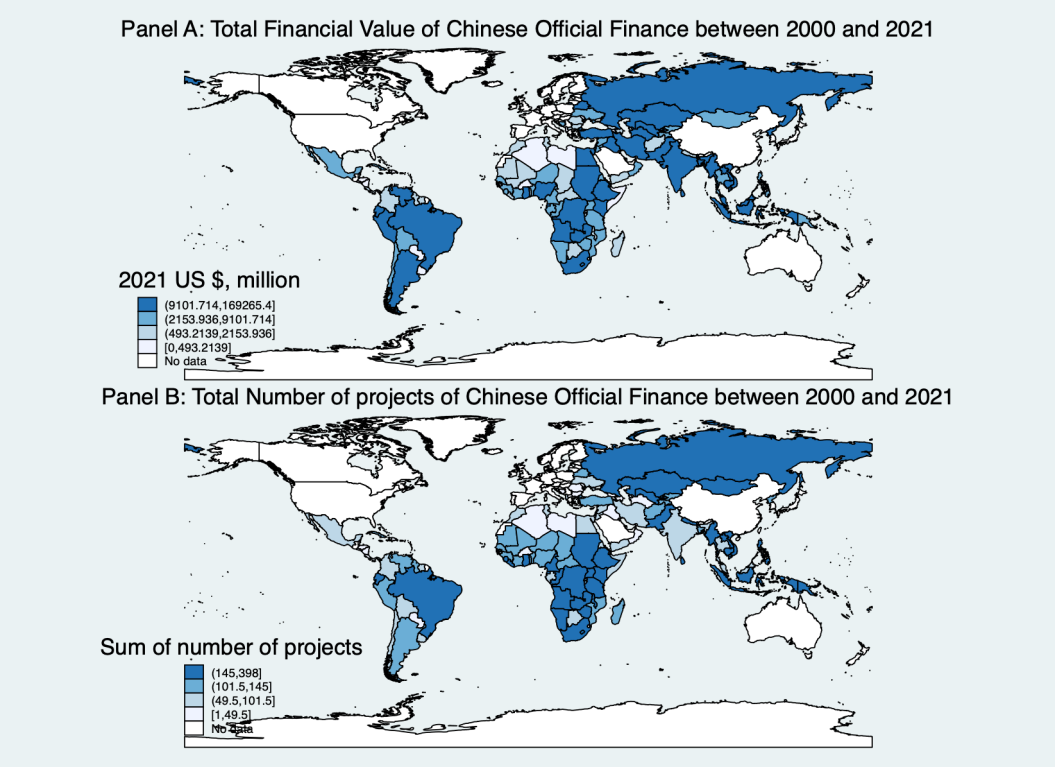
Yu, Run, Junning Cai, and PingSun Leung. "The normalized revealed comparative advantage index." *The Annals of Regional Science* 43 (2009): 267-282.

Figure 1. Chinese Official Foreign Aid by Category (2000-2021)



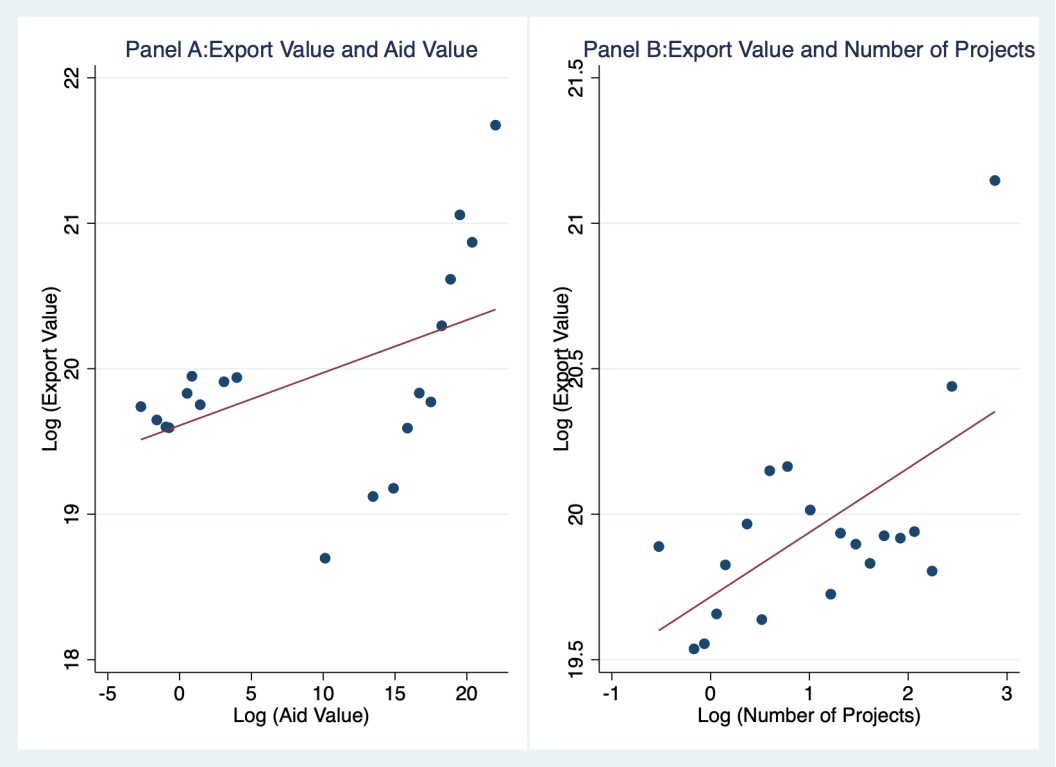
Notes: Panel A presents the total value of Chinese official foreign aid and the values of aid classified as official development assistance (ODA), other official finance (OOF), and unclassified official finance (Vague) from 2000-2021 in 2021 constant US dollars. Panel B shows the total count of Chinese official aid projects and the number of each category of Chinese official aid projects for the same period.

Figure 2. Geographical Distribution of Chinese Official Aid



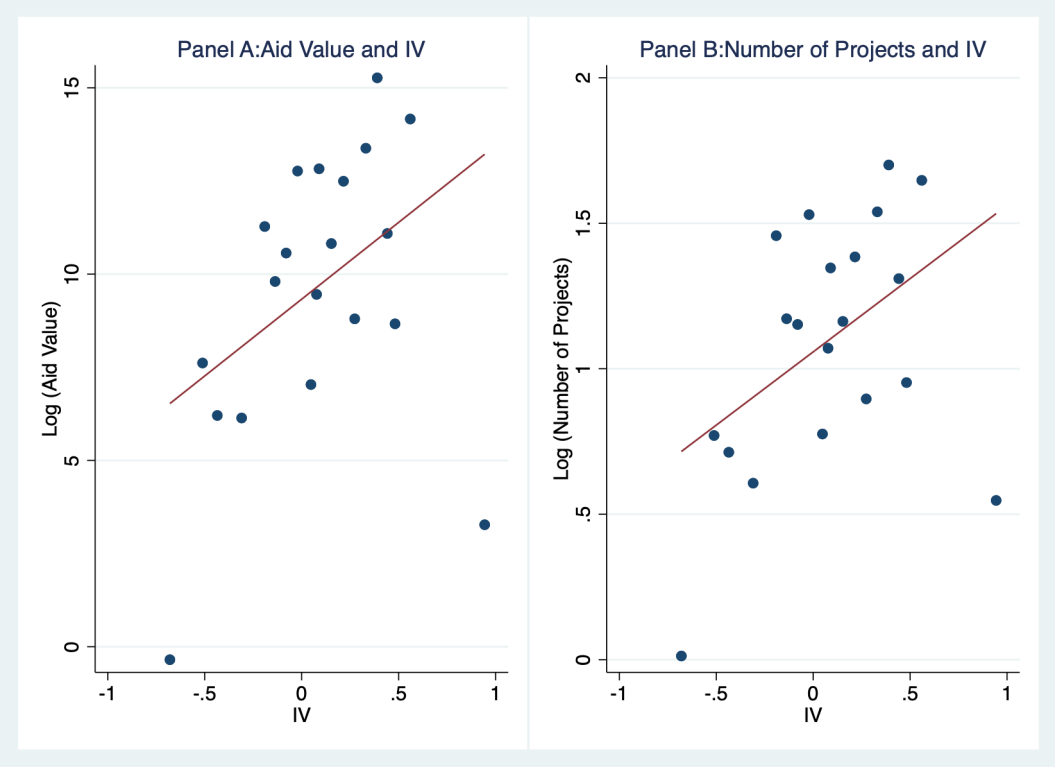
Notes: This figure depicts the global distribution of Chinese official foreign aid between 2000 and 2021. Panel A shows the distribution of total aid value, while Panel B illustrates the distribution of total number of projects.

Figure 3. Chinese Foreign Official Aid and the IV



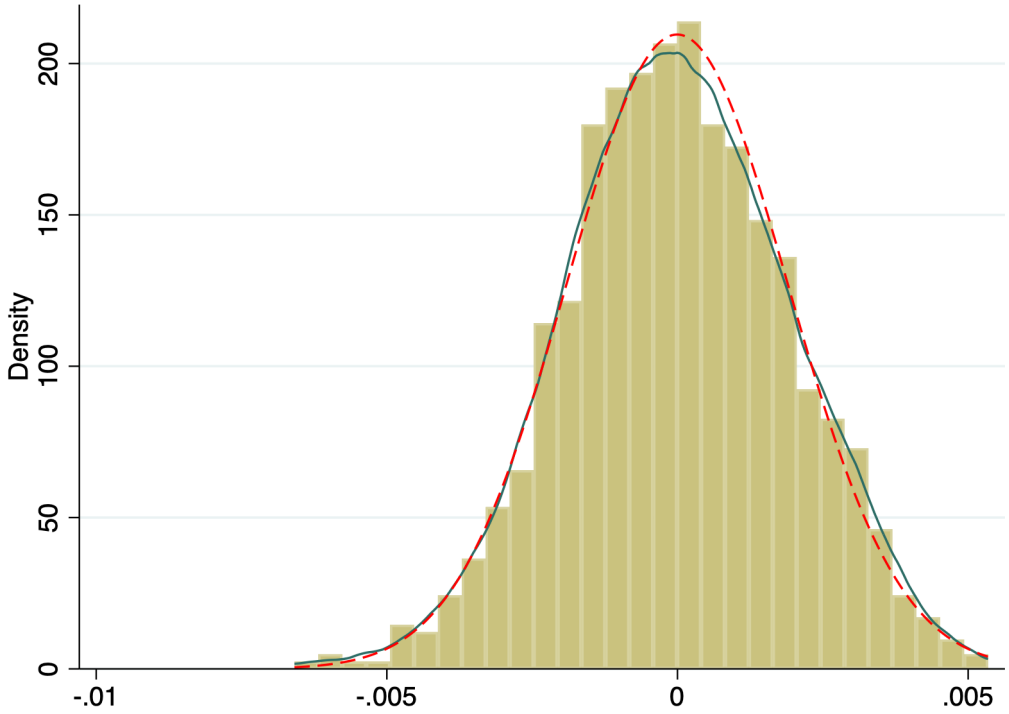
Notes: This figure shows binned scatter plots of Chinese official foreign aid against the IV. Chinese official foreign aid is measured by log aid value and log number of projects in Panels A and B, respectively. We obtain the plots by regressing measures of Chinese official foreign aid on the IV, while controlling for year fixed effects.

Figure 4. Chinese Foreign Official Aid and Exports to Recipient Country



Notes: This figure shows binned scatter plots illustrating the relationships between Chinese official foreign aid and China's exports to recipient countries. Chinese official foreign aid is measured by log aid value and log number of projects in Panels A and B, respectively. We obtain the plots by regressing log export value on measures of Chinese aid, while controlling for year fixed effect.

Figure 5. Distribution of Estimated Coefficients on Aid in Placebo Test



Notes: Figure 5 presents the distribution of coefficients on the log value of Chinese foreign official aid obtained from a placebo test. To conduct this test, we randomly reassigned the value of Chinese official foreign aid across countries and years, then re-estimated the baseline IV regression using this artificially generated data. We repeated this process 1000 times. The dash (red) line represents a normal distribution, while the solid line depicts the kernel distribution of the placebo coefficients.

Table 1. Summary Statistics of Key Variables

Variables	N	Mean	SD	Min	Max
Panel A: Country-level Variables					
Export value (log)	2,446	20.258	2.515	5.61	26.455
Aid value (log)	2,446	9.382	9.085	0	24.492
Number of projects (log)	2,446	1.066	1.012	0	4.466
Exchange rate (log)	2,446	-0.961	3.431	-9.29	27.828
GDP (log)	2,446	24.154	1.929	17.8	28.795
GDP growth rate	2,446	3.685	5.098	-54	43.480
Population (log)	2,446	15.601	2.014	8.38	21.065
Average tariff (log)	2,446	2.208	0.590	0	4.216
Panel B: Country-Industry level Variables					
Export value (log)	83,178	15.362	3.295	-2.08	25.739
Average tariff (log)	83,178	2.110	0.946	0	8.007
Panel C: Country-Product level Variables					
Export value (log)	4,874,367	10.735	3.170	-6.9	24.825
Tariff (log)	4,874,371	1.789	1.157	0	8.007

Notes: Panel A presents summary statistics for the main variables at the country level, Panel B provides the summary statistics of variables at the country-industry (ISIC Rev2, 3-digit) level, and Panel C presents summary statistics for variables at the country-product (HS1992, 6-digit) level.

Table 2. Baseline IV Regression Results

	(1) Baseline IV	(2) Alternative IV1	(3) Alternative IV2
Panel A: Second-stage results			
Log aid value	0.095***	0.094**	0.088*
	(3.122)	(2.469)	(1.730)
Log GDP	0.373	0.371	0.365
	(0.975)	(0.973)	(0.916)
GDP_growth	0.007	0.007	0.007
	(1.105)	(1.057)	(1.025)
Log Population size	0.781	0.786	0.807
	(1.004)	(1.014)	(0.986)
Exchange rate	0.040	0.037	0.025
	(0.220)	(0.195)	(0.136)
Log Tariff rate	-0.147	-0.146	-0.143
	(-1.129)	(-1.126)	(-1.108)
Panel B: First-stage results			
IV	1.830***	1.596***	1.719***
	(6.642)	(5.148)	(2.934)
No. of Obs.	2,153	2,153	2,153
Kleibergen-Paap F Statistic	44.121	26.506	8.606
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Notes: This table presents the baseline IV results and the robustness checks of using alternative IVs. All regressions include a constant term, and country and year fixed effects. Robust standard errors are clustered at the country level, and the corresponding t-statistics are provided in parentheses. The Kleibergen-Paap F statistics for the weak IV test are also reported.

Table 3. Robustness Checks

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Alternative measure of aid	Exclude buyer or supplier credit	Control aid from other donors	Drop top and bottom 1%	Exclude indirect effect	Exclude 2nd order indirect effect	Control growth ×country FE
Log no. of projects	0.851***						
	(3.109)						
Log aid value		0.096***	0.065**	0.085**	0.094***	0.085***	0.076***
		(3.078)	(2.385)	(2.497)	(3.062)	(2.844)	(2.938)
Log DAC value			0.022***				
			(3.419)				
Log IBRD-IDA value			-0.026*				
			(-1.948)				
Log UN value			-0.004				
			(-0.444)				
No. of Obs.	2,153	2,153	2,153	1,968	2,153	2,152	2,153
Kleibergen-Paap F Statistic	45.663	43.822	41.051	40.152	44.121	43.882	47.665
Country-year Level Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	No
China GDP growth×country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports the results of robustness checks. Column (1) uses the log number of projects as an alternative measure of Chinese aid. Column (2) excludes buyer's and supplier's credits in the calculation of aid value. Columns (3) controls for the log value of contributions from other donors. Column (4) excludes observations associated with the top or bottom 1 percentile of each variable. Column (5) excludes the export value of downstream sectors that use the six materials as primary inputs. Column (6) further excludes the export value of second-order downstream sectors. Column (7) additional controls for Chinese GDP growth×country fixed effects. All regressions include a constant term, country-year level controls, and year fixed effects. Columns (1) – (6) include country fixed effects, and column (7) include Chinese GDP growth×country fixed effects. Robust standard errors are clustered at the country level, and the corresponding t-statistics are provided in parentheses. The Kleibergen-Paap F statistics for the weak IV test are also reported.

Table 4. Trade Creation or Trade Diversion

	(1) Total imports from the world	(2) Imports from the ROW	(3) Imports from China/total imports
Log aid value	0.021**	0.017*	0.004*
	(1.995)	(1.731)	(1.974)
No. of Obs.	2,153	2,153	2,153
Kleibergen-Paap F Statistic	44.121	44.121	44.121
Country-year Level Controls	Yes	Yes	Yes
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Notes: This table investigates the effects of Chinese official foreign aid on recipient country's total imports and its composition. The dependent variables in columns (1) – (3) are the log of the recipient country's total import values from the world, the log of the recipient country's import value from all countries except China, and the ratio of the recipient country's import value from China to its total import value, respectively. All regressions include a constant term, country-year level controls, and year and country fixed effects. Robust standard errors are clustered at the country level, and the corresponding t-statistics are provided in parentheses. The Kleibergen-Paap F statistics for the weak IV test are also reported.

Table 5: Country-Industry Level Evidence on Mechanisms

	Imports from China				Imports from the ROW			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A								
Log aid value	0.075***	0.070***			0.017*	0.021*		
	(3.345)	(2.756)			(1.779)	(1.844)		
Log aid value×extfin			0.039***	0.027*			0.012***	0.015***
			(3.448)	(1.975)			(2.997)	(2.693)
Log aid value×contract			0.075***	0.058***			0.004	-0.004
			(4.668)	(3.211)			(0.625)	(-0.537)
Kleibergen-Paap F Statistic	47.733	26.516	21.426	11.418	47.731	26.512	21.426	11.418
Panel B								
Log no. of projects	0.694***	0.789**			0.157*	0.236*		
	(3.240)	(2.393)			(1.810)	(1.806)		
Log no. of projects×extfin			0.402***	0.375*			0.125***	0.205**
			(3.522)	(1.879)			(2.939)	(2.387)
Log no. of projects×contract			0.784***	0.797***			0.044	-0.055
			(4.528)	(2.752)			(0.667)	(-0.490)
Kleibergen-Paap F Statistic	41.232	15.566	15.880	5.026	41.233	15.566	15.880	5.026
No. of Obs.	71,186	71,186	71,178	71,178	71,185	71,185	71,177	71,177
Country-industry time trend	No	Yes	No	Yes	No	Yes	No	Yes
Country-level Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	No	Yes	Yes	No	No
Country×Ind FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country×Year FE	No	No	Yes	Yes	No	No	Yes	Yes

Note: This table presents the country-industry level IV regression results on mechanisms. Panels A and B use log aid value and log number of projects as the measure of Chinese official aid, respectively. Columns (1) - (4) report the effect on imports from China, while columns (5) - (8)

show the effect on imports from ROW. Columns (1), (2), (5) and (6) control for year and country-industry fixed effects, and the rest columns control for country-year and country-industry fixed effects. Columns (2), (4), (6), and (8) additionally control for country-industry time trend. All regressions include a constant term and country-year level controls. Robust standard errors are clustered at the country level, and the corresponding t-statistics are provided in parentheses. The Kleibergen-Paap F statistics for the weak IV test are also reported.

Table 6. Evidence on the Infrastructure Inputs Explanation

	Country-level (1)	Country-industry-level (2)	Country-product-level (3)	Upstream products (4)	Downstream products (5)
Panel A					
Log aid value	0.093***	0.068***	0.059***	0.048**	0.071***
	(3.028)	(3.073)	(2.836)	(2.376)	(3.287)
Kleibergen-Paap F Statistic	44.121	46.920	26.964	25.535	29.229
Panel B					
Log no. of projects	0.827***	0.635***	0.687**	0.567**	0.830***
	(3.023)	(2.989)	(2.464)	(2.129)	(2.793)
Kleibergen-Paap F Statistic	45.663	40.057	10.876	10.173	12.216
No. of Obs.	2,153	70,166	3,136,100	1,840,947	1,549,214
Country-level Controls	Yes	Yes	Yes	Yes	Yes
Country×Product FE	No	No	Yes	Yes	Yes
Country×Industry FE	No	Yes	No	No	No
Country FE	Yes	No	No	No	No
Year FE	Yes	Yes	Yes	Yes	Yes

Notes: The first three columns report results excluding the five largest inputs to the construction industry, as identified in the 2007 Chinese Input-Output tables at different levels of aggregation. Column (1) presents country-level analysis controlling for country and year fixed effects; Column (2) disaggregates to the country-industry level (ISIC 3-digit) and controls for country-industry and year fixed effects; Columns (3) shows country-product-level results (HS 6-digits) and controls for country-product fixed effects. Columns (4) and (5) also use country-product level data but further divide the sample by upstreamness (Chor et al., 2021). All regressions include a constant term. Robust standard errors are clustered at the country level, and the corresponding t-statistics are provided in parentheses. The Kleibergen-Paap F statistics for the weak IV test are also reported.

Table 7. Evidence on the Reduction of Transportation Cost Mechanism

	Value-to-weight ratio		Cif-to-fob ratio	
	(1)	(2)	(3)	(4)
Log aid value	0.054***		0.058***	
	(2.725)		(2.859)	
Log aid value×ratio	0.008	0.006	0.068	0.072
	(0.664)	(0.458)	(0.537)	(0.588)
No. of Obs.	2,131,270	2,131,265	3,396,605	3,396,598
Kleibergen-Paap F Statistic	12.978	728.104	13.676	37.850
Country-year Level Controls	Yes	Yes	Yes	Yes
Year FE	Yes	No	Yes	No
Country×Product FE	Yes	Yes	Yes	Yes
Country×Year FE	No	Yes	No	Yes

Notes: This table tests the reduction of transportation cost explanation by estimating equation (22) with the country-product level data. Columns (1) and (2) incorporate the interaction term between Chinese official aid and the value-to-weight ratio of the product, while columns (3) and (4) include the interaction term between Chinese official aid and the CIF-to-FOB ratio of the product. All regressions include a constant term. Columns (1) and (3) control for country-product fixed effects and year fixed effects, and columns (2) and (4) control for country-product fixed effects and country-year fixed effects. Robust standard errors are clustered at the country level, and the corresponding t-statistics are provided in parentheses. The Kleibergen-Paap F statistics for the weak IV test are also reported.

Table 8. Evidence on the Reduction of Information Frictions Mechanism

	Country-level evidence		Country-product-level evidence	
	(1)	(2)	(3)	(4)
Log aid value	-0.104	0.086*	0.110***	
	(-0.350)	(1.889)	(3.927)	
Log aid value×Distance	0.022			
	(0.705)			
Log aid value×Linguistic proximity		-0.040		
		(-0.463)		
Log aid value×RCA			-0.068***	-0.073***
			(-4.432)	(-4.540)
RCA			1.011***	1.068***
			(6.461)	(6.361)
No. of Obs.	2,101	1,974	1,898,514	1,898,479
Kleibergen-Paap F Statistic	17.510	17.262	10.588	24.296
Country-year Level Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	No
Country FE	Yes	Yes	No	No
Country×Product FE	No	No	Yes	Yes
Country×Year FE	No	No	No	Yes

Notes: This table tests the reduction of information friction explanation. Columns (1) and (2) provide evidence from country-level regressions. Column (1) includes the interaction term between aid and distance in equation (1), and columns (2) incorporate the interaction term between aid and linguistic proximity. Columns (3) and (4) provide evidence from country-product-level regressions. We include RCA and its interaction term with aid in equation (2). All regressions include a constant term. Columns (1) and (2) control for country and year fixed effects. Column (3) controls for country-product and year fixed effects, and column (4) controls for country-product fixed effects and country-year fixed effects. Robust standard errors are clustered at the country level, and the corresponding t-statistics are provided in parentheses. The Kleibergen-Paap F statistics for the weak IV test are also reported.

Table 9: Evidence on the Government Purchase Mechanism

	Country-level data	Country-industry level data			
	(1)	(2)	(3)	(4)	(5)
Log aid value	0.082	0.069***		0.072***	
	(1.252)	(3.038)		(3.144)	
Log aid value×gov_share	0.000				
	(0.049)				
Log aid value×high_purchase		-0.005	-0.007		
		(-0.450)	(-0.630)		
Log aid value×high_import				-0.017	-0.023**
				(-1.625)	(-2.120)
No. of Obs.	1,915	104,743	104,734	104,743	104,734
Kleibergen-Paap F Statistic	7.465	22.770	43.758	22.799	45.926
Country-year Level Controls	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	Yes	No
Country FE	Yes	No	No	No	No
Country×Ind FE	No	Yes	Yes	Yes	Yes
Country×Year FE	No	No	Yes	No	Yes

Notes: This table tests the government purchase hypothesis. Column (1) estimates equation (21) using country-level data and includes the interaction term with government expenditure share as an additional regressor. In columns (2) and (3), we include the interaction term between Chinese official finance and a dummy variable indicating an industry's government purchase share. In columns (4) and (5), we include the interaction term between Chinese official finance and a dummy variable indicating an industry' government import share. Columns (2) and (4) control for country-industry, and year fixed effects. Columns (3) and (5) control for country-year, and country-industry fixed effects. All regressions include a constant term. Robust standard errors are clustered at the country level, and the corresponding t-statistics are provided in parentheses. The Kleibergen-Paap F statistics for the weak IV test are also reported.

Online Appendix (Not for Publication)

Appendix A. Appendix Table

Table A1: Controlling for RCA

	(1)	(2)	(3)	(4)
Log aid value×extfin	0.038***	0.027*	0.034***	0.026*
	(3.345)	(1.907)	(2.843)	(1.748)
Log aid value×contract	0.072***	0.057***	0.078***	0.061***
	(4.566)	(3.036)	(4.712)	(3.164)
Log aid value×RCA			-0.074***	-0.057***
			(-5.082)	(-3.965)
RCA	0.159***	0.030	0.863***	0.575***
	(6.320)	(1.343)	(5.610)	(3.848)
Kleibergen-Paap F Statistic	20.139	10.635	15.043	7.037
Panel B				
Log number of projects×extfin	0.387***	0.379*	0.340***	0.406*
	(3.429)	(1.776)	(2.985)	(1.713)
Log number of projects×contract	0.738***	0.788**	0.724***	0.766***
	(4.524)	(2.593)	(4.656)	(2.682)
Log number of projects×RCA			-0.670***	-0.664***
			(-4.513)	(-3.323)
RCA	0.150***	0.033	0.861***	0.730***
	(5.782)	(1.474)	(5.041)	(3.307)
Kleibergen-Paap F Statistic	15.502	4.535	9.639	3.014
Control country-industry time trend	No	Yes	No	Yes
No. of Obs.	68809	68809	68809	68809
Country-year Level Controls	Yes	Yes	Yes	Yes
Country×Industry FE	Yes	Yes	Yes	Yes
Country×Year FE	Yes	Yes	Yes	Yes

Note: This table presents robustness check that augment the baseline specification with importer and Chinese industry-level RCA. Columns (1) and (2) include a dummy variable for RCA, which equals one if China has a revealed comparative advantage relative to the recipient country, and zero otherwise. Columns (3) and (4) add the interaction term between Chinese official finance and the RCA dummy. Columns (2) and (4) additionally control country-industry specific time trends. All regressions include a constant term and control for country-year and country-industry fixed effects. Robust standard errors are clustered at the country level, and the corresponding t-statistics are provided in parentheses. The Kleibergen-Paap F statistics for the weak IV test are also reported.

Appendix B. Variable Definitions and Data Sources

1. Country-level variables:

Export Value is the annual value of Chinese exports to recipient countries. We first exclude products related to iron, steel, cement, glass, timber, and aluminum by dropping all HS codes whose first two digits are 25, 26, 27, 44, 68, 70, 72, and 76. The data is then aggregated to the country level. Nominal export values are deflated to constant 2021 U.S. dollars using the recipient country's implicit price deflator. Finally, we take the natural logarithm of the deflated export value as our dependent variable. Source: UN Comtrade.

Financial Value is the annual amount of China's foreign assistance provided to each recipient country, measured in constant 2021 US dollars. Source: AidData.

Number of Projects is the total count of Chinese foreign assistance projects undertaken in each recipient country annually. Source: AidData.

GDP is the recipient country's real GDP each year, measured in constant 2021 US dollars. Source: UN Statistics.

GDP_d is the annual real GDP growth rate of recipient country. Source: UN Statistics.

Population is the total number of individuals of recipient country each year. Source: UN Statistics.

Exchange rate is the ratio of Chinese RMB to local currency of recipient country, measured annually. Source: UN Statistics.

Tariff is the simple average tariff across products, calculated at the country level for the recipient country each year. Source: WITS.

2. Country-industry level variables

Export Value is the annual value of Chinese export to recipient country at the country-industry (ISIC 3-digit) level each year. Using UN Comtrade data at the country-product (HS1992, 6-digit) level, we exclude products in the iron, steel, cement, glass, timber, and aluminum categories by dropping all HS codes whose first two digits are 25, 26, 27, 44, 68, 70, 72, and 76. The data is then aggregated to the country-industry level using the HS1992-ISIC concordance table from WITS. Nominal values are deflated to constant 2021 US dollars with recipient's implicit price deflator, and then natural logarithm is taken as the dependent variable. Source: UN Comtrade.

Tariff is recipient's import tariff rate at country-industry level each year. It is constructed as the simple average of product-level tariffs within industry. Source: WITS.

3. Country-product level variables

Export Value is the annual value of Chinese exports to a recipient country at the country-product level (six-digit, HS1992) for each year. We exclude products in the iron, steel, cement, glass, timber, and aluminum categories by dropping all HS codes whose first two digits are 25, 26, 27, 44, 68, 70, 72, and 76. Nominal values are deflated to constant 2021 US dollars with recipient's implicit price deflator, and then natural logarithm is taken as the dependent variable.

Source: UN Comtrade.

Tariff is recipient's import tariff at the HS 1992 6-digit product level for each year. Source: WITS.

4. Industry level variables

External Finance Dependence is defined as total assets minus retained earnings over total assets. Source: Fisman and Love (2003).

Contract intensity measures the fraction of differentiated inputs used production. Source: Nunn (2007).